

Power Scheduling for Coverage-Guided RTL Fuzzing

ECEN 689, Security Validation and Verification of Computing Systems

Team-Assert3: Lemuel Chineyemba, Vamsi Sree Sai Ganesh Kakumanu, Bhadradri Chaitanya Viriyala
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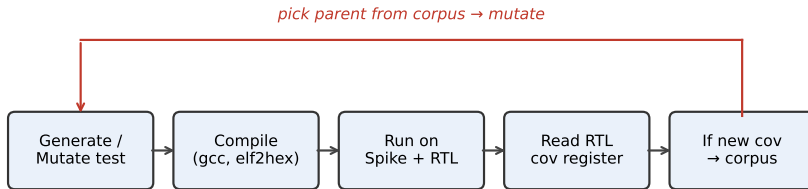
Department of ECE, Texas A&M University

Coverage Gains from Lightweight Scheduling

	Rocket	BOOM v1.3
DifuzzRTL artifact	83,003	485,055
Proposed method	87,212	510,410
Improvement	+5.07%	+5.23%

Coverage at 9-hour wall-clock; geometric mean across all runs.

DifuzzRTL Framework Overview



DifuzzRTL executes identical RISC-V tests on both a trusted reference simulator and the target hardware design. A coverage register within the hardware identifies inputs that trigger new behavior.

Baseline Parent Selection Mechanism

```
seed_si = random.choice(self.corpus)
```

Uniform selection across all seeds

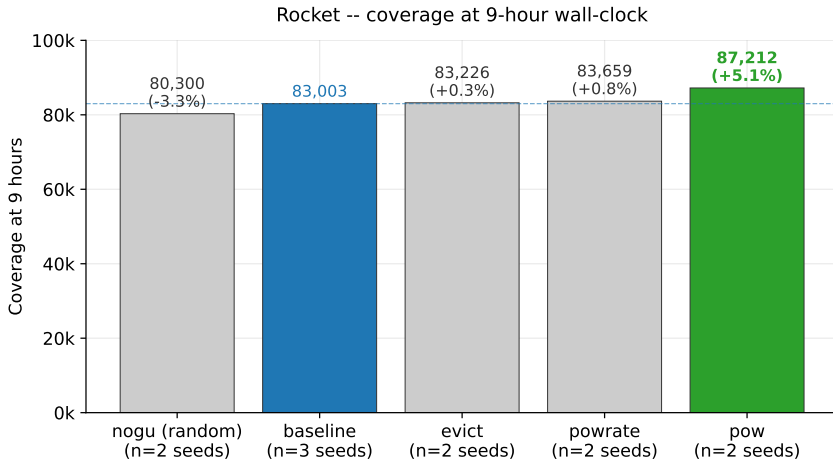
High-value seeds (frequent coverage gains) are treated identically to unproductive seeds.

Proposed Scheduling Strategies

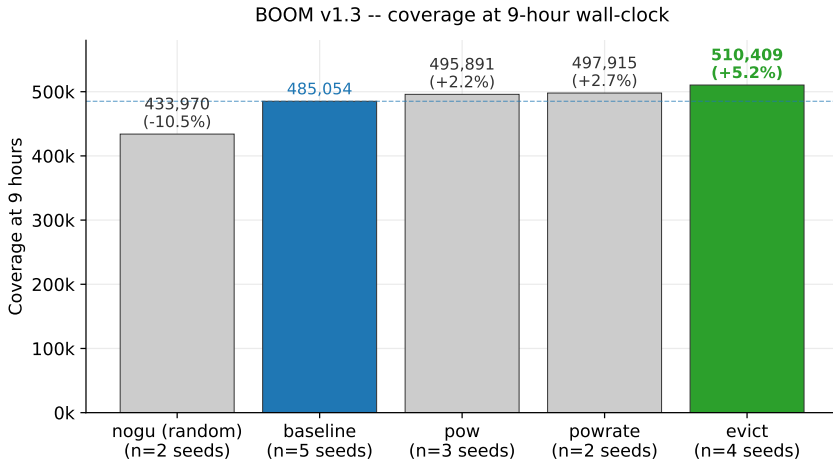
Variant	Selection Policy	Eviction Policy
baseline	Uniform random selection	FIFO
pow	Total reward count	FIFO
powrate	Smoothed success rate	FIFO
evict	Smoothed success rate	Adaptive (drop low-value seeds)

~30 lines of Python; controlled via DIFUZZ_SCHED environment variable. No recompilation required.

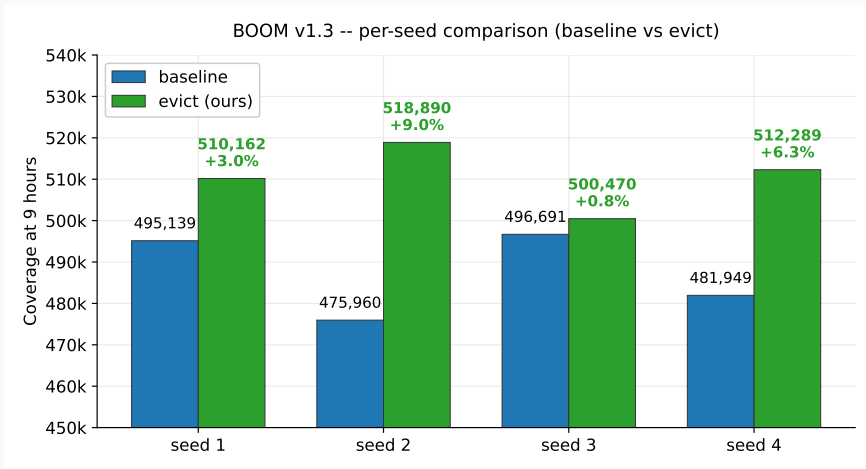
Rocket Core Results: pow Achieves +5.07%



BOOM v1.3 Results: *evict* Achieves +5.23%



Multi-Seed Evaluation on BOOM v1.3



Architecture-Dependent Scheduler Performance

Rocket

In-order, $\sim 90\text{k}$ states

Best: `pow`

Seed productivity is relatively uniform.

FIFO and adaptive eviction perform similarly.

Total-reward weighting is sufficient.

BOOM v1.3

Out-of-order, $\sim 500\text{k}$ states

Best: `evict`

A small subset of seeds dominates coverage gains.

FIFO prematurely discards high-value seeds.

Adaptive eviction preserves critical inputs.

Corpus saturation observed on both cores ($\sim 5,000$ – $9,800$ events vs. $1,000$ capacity); eviction policy is critical for larger designs.

Key Takeaways

Implementation: ~30 lines of Python

Overhead: No measurable runtime cost

Rocket: +5.07% coverage

BOOM v1.3: +5.23% coverage

Recommendation: Use evict for unknown targets